BETHANY HIGH SCHOOL, Sarjapur, Bangalore

WORKSHEET

CHAPTER: ATOMIC STRUCTURE AND CHEMICAL BONDING

Std – VIII

CHEMISTY

Question: 1

Choose the correct answers to these questions from the given options.

(Do not copy the question, write the correct answers only.)

(i) What type of bond forms between two non-metal atoms?

a) Ionic bond
b) Covalent bond
c) Metallic bond
d) Hydrogen bond

(ii) In an ionic bond, electrons are:

a) Shared between atoms
b) Transferred from one atom to another
c) Present in a "sea" of electrons
d) Not involved in bonding

(iii) The electrons in an atom are found in specific:

(I) Shells or energy levels
(II) Nucleon
(III) Nucleus

a) Only (I)
b) Only (II)
c) Both (II) and (III)
d) Only (III)

(iv) Water molecules are held together by which type of bond.

a) Ionic bond
b) Covalent bond
c) Metallic bond
d) Hydrogen bond

(v) Isotopes of an element have the same number of:

a) Protons but different numbers of neutrons
b) Protons and electrons
c) Neutrons but different numbers of protons
d) Electrons but different numbers of protons

(vi) The atomic number of an element is determined by the number of:

a) Protons in the nucleus
b) Neutrons in the nucleus
c) Electrons in the nucleus
d) Protons and neutrons combined

(vii) Which of the following element typically forms covalent bond?

a) Sodium
b) Lithium
c) Carbon
d) Potassium

(viii) If 'n' is the serial number of the shell, then maximum number of electrons can be determined by the formula:
(a) $2n^2$
(b) 6n
(c) $3n^2$
(d) 2(n+1)

(ix) What is/are the total number loan pair of electrons present in a methane (CH_4) molecule?
a) 2
b) 4
c) 8
d) 0

Question: 2

(i) Fill in the blanks:

a) The outermost shell of an atom is known as the _____shell.

b) Atoms combine to form molecules through _____bonds.

c) The negatively charged subatomic particles in an atom are called _____.

d) The valency of an element determines its ability to _____with other elements.

e) An atom becomes an ion when it gains or loses _____.

(ii) Representation of an element is $_{13}^{27}Z$

Answer the following question:

a) What is the valency of Z?

b) What is the role of valence electrons in an atom?

c) How do valence electrons influence an atom's bonding behaviour?

d) State the formula of the compound formed by Z & nitrate.

e) What kind of bonding is present in the above compound?

Question: 3

(i) Give the orbital diagram for the formation of Water by three steps method. [Atomic number of H=1, O=8]

(ii) Analogy-based question.

a) Na^+ : Cation :: Cl^- : _____

b) K^+ and F^- : Ionic bond :: H and F : _____

(iii) Define the following term:

a) Valence electron

b) Duplet rules

c) Atomic number

d) Isotope

e) Octate rules

f) Nucleon

(iv) Give a reason for each of the following:

a) Isotopes have similar chemical properties.

b) $MgCl_2$ an example of electrovalent bond, CCl_4 as a covalent bond.

Question: 4

Answer the following question:

(i) The atomic number of sodium is 11. What is the number of electrons in Na^+?

(ii) An element X has 2 electrons in its M shell, it forms bond with an element Y which has 7 electrons in its 3rd orbit.

a) Write the formula of the compound form.

b) Which nearest inert gas electronic configuration will element X&Y acquire.

c) Show by orbital diagram the formation of compound between X&Y

(iii) An element M burns in oxygen to form an ionic compound MO. Write the formula of the compound form if this element is made to combine with:

a) sulphide

b) nitrite

(iv) State the type of covalent bonding in the following molecules:

a) Nitrogen

b) Methane

c) Hydrogen